

Covariance Geometry of Linearized Quantum Erasure Correction

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Abstract

A fixed-dimensional quantum code is a point of a Grassmannian, and exact erasure correction cuts out a Knill–Laflamme (KL) zero locus inside that Grassmannian. We identify the first-order normal geometry of this zero locus. For a code that exactly corrects erasure of subsystem A , with constant erased marginal σ_A , the vertical differential of the KL section obeys

$$\boxed{d^v s_A (d^v s_A)^* = 4 \operatorname{Cov}_{\sigma_A} \otimes I_{\operatorname{Herm}_0(L)}.$$

Thus the nonzero singular spectrum of the linearized exact-QEC constraints is determined by the covariance spectrum of the erased marginal. This identity has three immediate consequences. First, full-rank exact erasure codes form a smooth KL zero locus of the expected codimension. Second, a quadratic mean-square leakage functional is the squared norm of the KL section up to the universal Haar factor $K(K+1)$; hence it is Morse–Bott along the full-rank exact locus and its normal Hessian is covariance-controlled. Third, for qubit erasure the reduced normal condition number is exactly the reciprocal of the common qubit–rest concurrence, $\kappa_{\text{red}} = 1/C_A$. For simultaneous erasure constraints we derive cross-Gram blocks and use exact symbolic Jacobian calculations to show infinitesimal rigidity modulo local physical unitaries for the $[[4, 2, 2]]$ and $[[5, 1, 3]]$ codes. We also state the corresponding covariance normal form for scalar-compressed observable algebras.

1 Introduction

A K -dimensional quantum code in an N -dimensional physical Hilbert space is naturally a point of the Grassmannian $\operatorname{Gr}(K, N)$. This geometric viewpoint is established: fixed-dimensional code spaces admit tautological and frame-bundle descriptions, higher-rank numerical ranges encode scalar-compression constraints, and exact quantum error correction produces structured families of error-image subspaces [1, 2, 3, 4]. Recent work has also optimized code subspaces directly on Stiefel or Grassmann manifolds and studied global compression and variance landscapes of exact codes [7, 8]. Covariance and moment-map structures are likewise standard ingredients in geometric quantum mechanics and entanglement theory [9, 10]. The question here is narrower: what is the *local normal geometry* of the exact-QEC equations at a code that already corrects a specified erasure?

Let $A \cong \mathbb{C}^a$, $B \cong \mathbb{C}^b$, and let $\mathcal{C} \subset A \otimes B$ be a K -dimensional code with projector P . Erasure of A is exactly correctable iff every local observable on A compresses to a scalar on the code,

$$P(T \otimes I_B)P = \tau(T)P. \tag{1}$$

Equivalently, every logical state has the same erased marginal σ_A [5, 6]. These equations define a natural vector-bundle section over the Grassmannian. Our main result computes the Gram operator of its vertical differential exactly.

At an exact code,

$$\boxed{d^v s_A (d^v s_A)^* = 4 \text{Cov}_{\sigma_A} \otimes I_{\text{Herm}_0(L)}}. \quad (2)$$

The covariance operator is taken on traceless Hermitian observables of the erased subsystem. Therefore, after quotienting tangent directions along the exact-code locus, the entire first-order stiffness spectrum of the KL constraints is a covariance spectrum of σ_A ; logical dimension changes only multiplicities.

The same operator reappears in the Hessian of a mean-square complementary-channel leakage functional. For a qubit erasure, the covariance spectrum is elementary and its smallest nonzero eigenvalue is controlled by concurrence, yielding $\kappa_{\text{red}}(d^v s_A) = 1/C_A$. For several correctable erasure regions, diagonal Gram blocks retain the covariance form while off-diagonal blocks are compressed cross-correlations. This gives a deformation-theoretic language in which local physical-unitary tangent directions lie in the kernel of the stacked KL Jacobian.

Novelty boundary. Grassmannian code geometry, covariance tensors, moment maps, and scalar-compression formulations of QEC are not new. The contribution claimed here is the exact local identity in Eq. (2), its normal-Hessian and concurrence consequences, and the simultaneous-erasure cross-Gram formula in this KL setting.

2 Preliminaries and the KL section

Let $L \cong \mathbb{C}^K$ be the logical Hilbert space and choose an encoding isometry

$$V : L \rightarrow A \otimes B, \quad V^\dagger V = I_L, \quad VV^\dagger = P.$$

Two frames V and VU , $U \in U(K)$, define the same code. For a finite-dimensional Hilbert space X , write

$$\text{Herm}_0(X) = \{T = T^\dagger : \text{Tr } T = 0\},$$

with Hilbert–Schmidt inner product $\langle S, T \rangle = \text{Tr}(ST)$. For $Y \in \text{Herm}(L)$ set

$$\text{tf}(Y) = Y - \frac{\text{Tr } Y}{K} I_L.$$

The real tangent space of the complex Grassmannian is identified with $\text{Hom}_{\mathbb{C}}(\mathcal{C}, \mathcal{C}^\perp)$. In an encoding frame we represent a tangent vector by $X : L \rightarrow \mathcal{C}^\perp$ satisfying $V^\dagger X = 0$ and use metric $g(X, Y) = \Re \text{Tr}(X^\dagger Y)$.

Let $\mathcal{S} \rightarrow \text{Gr}(K, A \otimes B)$ denote the tautological bundle. Define

$$s_A(\mathcal{C})(T) = \text{tf}(P(T \otimes I_B)P|_{\mathcal{C}}), \quad T \in \text{Herm}_0(A). \quad (3)$$

This is a section of $\text{Herm}_0(A)^* \otimes \text{Herm}_0(\mathcal{S})$.

Proposition 2.1 (Exact erasure codes are the zero locus). *A code $\mathcal{C}$ exactly corrects erasure of A iff $s_A(\mathcal{C}) = 0$. Thus*

$$\mathcal{M}_A := s_A^{-1}(0) \subset \text{Gr}(K, ab)$$

is the exact erasure-correcting locus.

Proof. If $s_A(\mathcal{C}) = 0$, every traceless Hermitian T has scalar compression. The identity does trivially, and complex linear combinations of Hermitian operators span $\text{End}(A)$, so the full KL condition follows. The converse is immediate. $\square$

For an exact code define the constant erased marginal

$$\sigma_A = \frac{1}{K} \text{Tr}_B P. \quad (4)$$

Then $V^\dagger(T \otimes I)V = \text{Tr}(\sigma_A T)I_L$.

2.1 Fixed-rank zero-locus geometry

Let $\mathcal{M}_A^{(r)}$ denote the subset for which $\text{rank } \sigma_A = r$. The no-hiding structure gives a useful parameterization. If $S_\sigma = \text{supp } \sigma$, a canonical purification $\Omega_\sigma \in A \otimes S_\sigma$ and an isometry $W : S_\sigma \otimes L \rightarrow B$ define

$$V_{\sigma, W}\psi = (I_A \otimes W)(\Omega_\sigma \otimes \psi). \quad (5)$$

Quotienting the free right action of $U(K)$ on the logical frame gives the code space.

Proposition 2.2 (Fixed-rank stratum). *If $rK \leq b$, then $\mathcal{M}_A^{(r)}$ is smooth with*

$$\dim_{\mathbb{R}} \mathcal{M}_A^{(r)} = (2ar - r^2 - 1) + (2brK - r^2K^2 - K^2). \quad (6)$$

For full rank $r = a$,

$$\dim_{\mathbb{R}} \mathcal{M}_A^{(a)} = 2K(ab - K) - (a^2 - 1)(K^2 - 1). \quad (7)$$

Proof. Rank- r density matrices on A form a real manifold of dimension $2ar - r^2 - 1$. For fixed support, the isometries $S_\sigma \otimes L \rightarrow B$ form a complex Stiefel manifold of real dimension $2brK - r^2K^2$. Quotienting the free $U(K)$ logical-frame action subtracts K^2 dimensions. $\square$

3 Covariance normal form of the linearized KL constraints

At a zero of a vector-bundle section, its vertical differential is canonical. In the horizontal frame above it is obtained by differentiating the fiber coordinate of Eq. (3).

Lemma 3.1 (Vertical differential). *At an exact code,*

$$(\text{d}^V s_A)[X](T) = \text{tf}\left(X^\dagger M_T V + V^\dagger M_T X\right), \quad M_T = T \otimes I_B. \quad (8)$$

Under the Hilbert–Schmidt identification of the section fiber with $\text{Herm}_0(A) \otimes_{\mathbb{R}} \text{Herm}_0(L)$,

$$(\text{d}^V s_A)^*(T \otimes L_0) = 2P^\perp M_T V L_0. \quad (9)$$

Proof. Differentiate $\text{tf}(V(t)^\dagger M_T V(t))$ along a horizontal lift $\dot{V}(0) = X$. Pairing the result with $L_0 \in \text{Herm}_0(L)$ gives twice the real part of $\text{Tr}[X^\dagger M_T V L_0]$. The component parallel to the code is $V \text{Tr}(\sigma_A T) L_0$, which is orthogonal to horizontal tangent vectors, leaving Eq. (9). $\square$

Define the symmetrized covariance form on $\text{Herm}_0(A)$ by

$$\text{Cov}_{\sigma_A}(S, T) = \frac{1}{2} \text{Tr}(\sigma_A \{S, T\}) - \text{Tr}(\sigma_A S) \text{Tr}(\sigma_A T). \quad (10)$$

Let Cov_{σ_A} be the self-adjoint operator representing this form.

Theorem 3.2 (Covariance normal form). *At every exact A -erasure code,*

$$\boxed{\mathrm{d}^{\mathrm{v}}s_A(\mathrm{d}^{\mathrm{v}}s_A)^* = 4 \operatorname{Cov}_{\sigma_A} \otimes I_{\operatorname{Herm}_0(L)}} \quad (11)$$

If ν_j are the covariance eigenvalues, the nonzero singular values of $\mathrm{d}^{\mathrm{v}}s_A$ are $2\sqrt{\nu_j}$, each repeated $K^2 - 1$ times.

Proof. Apply the composition to $T \otimes L_0$. By Eq. (9), set $X = 2P^\perp M_T V L_0$. For $S \in \operatorname{Herm}_0(A)$,

$$(\mathrm{d}^{\mathrm{v}}s_A)[X](S) = 2 \operatorname{tf} \left[L_0 V^\dagger M_T P^\perp M_S V + V^\dagger M_S P^\perp M_T V L_0 \right].$$

Exactness implies

$$V^\dagger M_T M_S V = \operatorname{Tr}(\sigma_A T S) I_L, \quad V^\dagger M_T P M_S V = \operatorname{Tr}(\sigma_A T) \operatorname{Tr}(\sigma_A S) I_L.$$

The second summand is the Hermitian conjugate of the first. Since L_0 is traceless Hermitian, the result is $4 \operatorname{Cov}_{\sigma_A}(S, T) L_0$. $\square$

Lemma 3.3 (Covariance kernel). *If $r = \operatorname{rank} \sigma_A$, then*

$$\dim_{\mathbb{R}} \ker \operatorname{Cov}_{\sigma_A} = (a - r)^2, \quad \operatorname{rank}_{\mathbb{R}} \operatorname{Cov}_{\sigma_A} = 2ar - r^2 - 1. \quad (12)$$

Proof. For traceless Hermitian T and $m = \operatorname{Tr}(\sigma_A T)$,

$$\operatorname{Cov}_{\sigma_A}(T, T) = \operatorname{Tr}[\sigma_A(T - mI)^2].$$

Writing $S = \operatorname{supp} \sigma_A$, this vanishes iff $T = mI_S \oplus D$ with no off-diagonal block between S and $S^\perp$, subject to the single global trace constraint. The free dimension is $(a - r)^2$. $\square$

Corollary 3.4 (Smooth full-rank locus). *At an exact code with full-rank erased marginal,*

$$\operatorname{rank}_{\mathbb{R}} \mathrm{d}^{\mathrm{v}}s_A = (a^2 - 1)(K^2 - 1). \quad (13)$$

In particular, $\mathrm{d}^{\mathrm{v}}s_A$ is surjective. Hence the full-rank exact locus is a smooth submanifold of codimension $(a^2 - 1)(K^2 - 1)$, in agreement with Eq. (7).

Corollary 3.5 (Normal coordinates). *If $\sigma_A > 0$ and $N_{\mathcal{C}} = (\ker \mathrm{d}^{\mathrm{v}}s_A)^\perp$ is the Riemannian normal space to the exact-code locus, there exists an orthogonal isomorphism*

$$U_{\mathcal{C}} : N_{\mathcal{C}} \rightarrow \operatorname{Herm}_0(A) \otimes_{\mathbb{R}} \operatorname{Herm}_0(L)$$

such that

$$\mathrm{d}^{\mathrm{v}}s_A|_{N_{\mathcal{C}}} = 2(\operatorname{Cov}_{\sigma_A}^{1/2} \otimes I) U_{\mathcal{C}}. \quad (14)$$

4 Leakage Hessian from covariance

For an arbitrary code, define its average erased marginal by Eq. (4). For a Haar-random logical pure state $|\psi\rangle \in \mathcal{C}$, let $\rho_A(\psi) = \operatorname{Tr}_B |\psi\rangle\langle\psi|$ and define

$$\mathcal{L}_A(\mathcal{C}) = \mathbb{E}_{\psi} \|\rho_A(\psi) - \sigma_A(\mathcal{C})\|_2^2. \quad (15)$$

The functional vanishes exactly on the erasure-correcting locus.

Theorem 4.1 (Exact leakage identity). *For every K -dimensional code,*

$$\boxed{\mathcal{L}_A(\mathcal{C}) = \frac{\|s_A(\mathcal{C})\|^2}{K(K+1)}}. \quad (16)$$

Proof. Choose an orthonormal basis $\{T_\mu\}$ of $\text{Herm}_0(A)$ and set $B_\mu = \text{tf}(V^\dagger(T_\mu \otimes I)V)$. Parseval gives

$$\|\rho_A(\psi) - \sigma_A\|_2^2 = \sum_\mu \langle \psi | B_\mu | \psi \rangle^2.$$

For traceless Hermitian B , the Haar second moment is $\mathbb{E}\langle \psi | B | \psi \rangle^2 = \text{Tr}(B^2)/[K(K+1)]$. Summing over μ yields Eq. (16). $\square$

Corollary 4.2 (Morse–Bott normal Hessian). *At a full-rank exact code, $\mathcal{L}_A$ is Morse–Bott along $\mathcal{M}_A^{(a)}$. On normal directions,*

$$\text{Hess } \mathcal{L}_A = \frac{2}{K(K+1)} (\text{d}^\vee s_A)^* \text{d}^\vee s_A. \quad (17)$$

Consequently its nonzero Hessian eigenvalues are

$$\frac{8\nu_j}{K(K+1)}, \quad (18)$$

with covariance multiplicities inherited from Eq. (11).

5 Qubit erasure and concurrence

Let $a = 2$ and let the eigenvalues of σ_A be $\lambda_\pm$. Define the common qubit–rest concurrence of the exact code by

$$C_A = 2\sqrt{\det \sigma_A} = 2\sqrt{\lambda_+ \lambda_-}.$$

For a Hilbert–Schmidt orthonormal Pauli basis of $\text{Herm}_0(\mathbb{C}^2)$, direct diagonalization of Eq. (10) gives covariance eigenvalues

$$\frac{1}{2}, \quad \frac{1}{2}, \quad \frac{C_A^2}{2}.$$

Definition 5.1. For a linear map D with nontrivial kernel, define the reduced condition number by

$$\kappa_{\text{red}}(D) = \frac{\sigma_{\max}^+(D)}{\sigma_{\min}^+(D)}.$$

Corollary 5.2 (Concurrence controls normal conditioning). *For a full-rank exact qubit-erasure code, the nonzero singular values of $\text{d}^\vee s_A$ are*

$$\sqrt{2}, \quad \sqrt{2}, \quad \sqrt{2} C_A, \quad (19)$$

each repeated $K^2 - 1$ times. Hence

$$\boxed{\kappa_{\text{red}}(\text{d}^\vee s_A) = \frac{1}{C_A}}. \quad (20)$$

The reduced condition number of the normal Hessian is $1/C_A^2$.

At $\sigma_A = I/2$, $C_A = 1$ and the normal KL geometry is isotropic. As $C_A \rightarrow 0$, one singular sector softens as the full-rank stratum approaches its rank-deficient boundary. For a pure state in $\mathbb{C}^2 \otimes \mathbb{C}^b$, if r_{FS} denotes Fubini–Study distance to the Segre variety along the Schmidt geodesic, $C_A = \sin(2r_{\text{FS}})$; the point here is that the same quantity controls the KL normal condition number.

6 Simultaneous erasure constraints and infinitesimal rigidity

Suppose a family $\mathcal{A}$ of erased regions must all be correctable. Stack the KL sections,

$$S_{\mathcal{A}} = \bigoplus_{R \in \mathcal{A}} s_R. \quad (21)$$

The first-order KL-preserving deformations at $\mathcal{C}$ are $\ker(d^{\vee} S_{\mathcal{A}})$. Tangents generated by local physical unitaries lie in this kernel. We call the code infinitesimally rigid modulo local physical unitaries when they exhaust it.

6.1 Cross-Gram blocks

Let R, S be separately correctable regions. For Hermitian observables A on R and B on S , embedded into the full physical algebra, write

$$V^{\dagger} A V = \omega_R(A) I_L, \quad V^{\dagger} B V = \omega_S(B) I_L,$$

and define

$$C_{AB} = V^{\dagger} A B V - \omega_R(A) \omega_S(B) I_L.$$

Proposition 6.1 (Cross-Gram formula). *At a code that corrects R and S separately,*

$$(d^{\vee} s_S (d^{\vee} s_R)^*)(A \otimes L)(B) = 2 \operatorname{tf}(L C_{AB} + C_{BA} L). \quad (22)$$

If $R \cup S$ is itself correctable, this reduces to

$$(d^{\vee} s_S (d^{\vee} s_R)^*)(A \otimes L)(B) = 4 \operatorname{Cov}_{\sigma_{R \cup S}}(A, B) L. \quad (23)$$

Proof. Insert $(d^{\vee} s_R)^*(A \otimes L) = 2 P^{\perp} A V L$ into the differential of s_S . The two terms are the compressions of BA and AB after subtracting the product of their scalar compressions. If the union is correctable, both products compress to scalars determined by the union marginal, yielding the covariance form. $\square$

6.2 Two exact rigidity calculations

The following ranks are obtained by exact symbolic row reduction over algebraic/rational entries, so no floating-point rank threshold enters the statements.

Proposition 6.2 (Four-qubit distance-two code). *For the standard $[[4, 2, 2]]$ code under all single-qubit erasure constraints, the stacked KL Jacobian has real rank 84 on the 96-dimensional Grassmann tangent space. Its kernel has dimension 12, equal to the dimension of the tangent orbit generated by the twelve one-qubit Pauli Hamiltonians. Hence the code is infinitesimally rigid modulo local physical unitaries.*

Proposition 6.3 (Five-qubit perfect code). *For the standard five-qubit perfect code under all two-qubit erasure constraints, the stacked KL Jacobian has real rank 105 on a 120-dimensional tangent space. Its 15-dimensional kernel equals the tangent orbit generated by the fifteen one-qubit Pauli Hamiltonians. Hence the code is infinitesimally rigid modulo local physical unitaries.*

These are first-order statements only; no global rigidity follows without controlling higher-order integration of the KL-preserving directions.

7 Scalar-compressed observable algebras

The covariance identity is not specific to a tensor-factor erasure. Let $\mathfrak{A} \subseteq \text{End}(\mathcal{H})$ be a finite-dimensional unital $*$ -subalgebra and let $\mathfrak{A}_{\text{sa},0}$ be its traceless Hermitian part. A rank- K code projector P is a scalar compression for $\mathfrak{A}$ if

$$PAP = \omega(A)P, \quad A \in \mathfrak{A},$$

for a state ω on the algebra. Define

$$s_{\mathfrak{A}}(\mathcal{C})(A) = \text{tf}(PAP|_{\mathcal{C}}).$$

Theorem 7.1 (Algebraic covariance normal form). *At every scalar-compressing code,*

$$\boxed{\text{d}^{\vee}s_{\mathfrak{A}}(\text{d}^{\vee}s_{\mathfrak{A}})^* = 4 \text{Cov}_{\omega} \otimes I_{\text{Herm}_0(L)}} \quad (24)$$

If ω is faithful on $\mathfrak{A}$, the scalar-compression locus is smooth at the code of real codimension $(\dim_{\mathbb{C}} \mathfrak{A} - 1)(K^2 - 1)$.

Proof. The differential and adjoint are identical to Eqs. (8)–(9) with M_T replaced by $A \in \mathfrak{A}$. The composition produces compressions of AB and BA ; because $\mathfrak{A}$ is an algebra these remain inside $\mathfrak{A}$ and therefore compress to the scalar values $\omega(AB)$ and $\omega(BA)$. The resulting symmetrization is exactly the covariance form. $\square$

8 Discussion

Equation (11) supplies a local spectral normal form for exact erasure correction. It separates global classification of exact codes from local stiffness of the KL equations at a fixed code. The erased marginal alone determines the nonzero normal singular spectrum; the logical dimension appears only through multiplicity. The same spectrum controls the quadratic leakage Hessian and, for qubit erasure, admits the closed concurrence formula.

Several boundaries are important. First, the leakage functional in Eq. (15) is chosen because it is exactly quadratic in the KL section; it is not asserted to be the unique operational metric for approximate QEC. Second, covariance is not itself a new object in quantum geometry. The content is the specific Gram identity for the KL scalar-compression section. Third, the stacked-Jacobian calculations certify only infinitesimal rigidity. Finally, rank-deficient erased marginals are singular strata of the full-rank zero-locus geometry; the covariance kernel formula gives their first-order rank loss, but their global stratified geometry can contain additional structure.

A natural next problem is to replace exact scalar compression by an operational approximate-QEC distance and determine which part of the covariance normal form survives after recovery optimization. Another is to classify when stacked cross-Gram blocks force infinitesimal rigidity for code families rather than isolated examples.

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